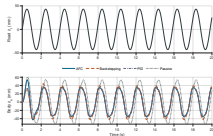

Approximation-Free Active Suspension Control

Na et al. (2022): paper analysis and MATLAB/Simulink reproduction

Advanced Control Theory course project

30 September 2026



Research question

Objective

Reduce body vibration under road disturbance while keeping a prescribed transient bound and feasible hydraulic command.

Paper method

Three-stage approximation-free control (AFC), with a shrinking prescribed-performance function (PPF) and transformed errors.

Our reproduction question

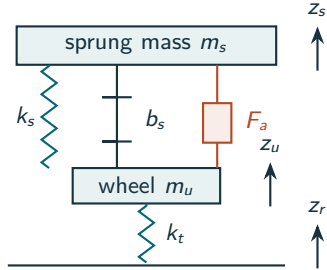
What can we verify from published equations and road cases when the original rig constants and raw signals are unavailable?

Evidence boundary

The figures here are a representative simulation, not a digitized or reconstructed hardware experiment.

Source paper: J. Na et al., IEEE TSMC: Systems 52(8), 4714–4726, 2022. DOI: 10.1109/TSMC.2021.3103807.

Physical system and model



$$F_s = k_s(z_s - z_u) + k_{sn}(z_s - z_u)^3$$

$$F_d = b_s(\dot{z}_s - \dot{z}_u)$$

$$m_s \ddot{z}_s = -F_s - F_d + F_a$$

$$m_u \ddot{z}_u = F_s + F_d - F_t - F_b - F_a$$

$$F_a = AP_L, \quad x_5 = \kappa P_L$$

Original schematic and model equations; nonlinear spring and tire damping are set to zero in the numerical experiment.

Prescribed performance transforms a constraint

$$\begin{aligned}\mu_i(t) &= (\mu_{i0} - \mu_{i\infty})e^{-\alpha_i t} + \mu_{i\infty} \\ -\delta\mu_i(t) &< e_i(t) < \bar{\delta}\mu_i(t) \\ \zeta_i &= \frac{e_i}{\mu_i}, \quad \epsilon_i = \frac{1}{2} \ln \left(\frac{\delta + \zeta_i}{\bar{\delta} - \zeta_i} \right)\end{aligned}$$

μ_{i0}	initial admissible error
α_i	contraction speed
$\mu_{i\infty}$	long-run error allowance

A bounded transformed error is the route to staying inside the shrinking envelope.

Approximation-free recursive law

$$e_1 = x_1 \longrightarrow u_1 = -k_1 \epsilon_1$$

$$e_2 = x_2 - u_1 \longrightarrow u_2 = -k_2 \epsilon_2$$

$$e_3 = x_5 - u_2 \longrightarrow u = -k_3 \epsilon_3$$

Case A channel	μ_{i0}	$\mu_{i\infty}$	α_i	k_i
Displacement	0.2	0.018	3	25.5
Virtual velocity	110	90	2	12
Pressure state	100	80	2	216

Case A values: Na et al. Symmetric $\delta = \bar{\delta} = 1$ is a project inference, not a published numeric setting.

What the paper actually reports

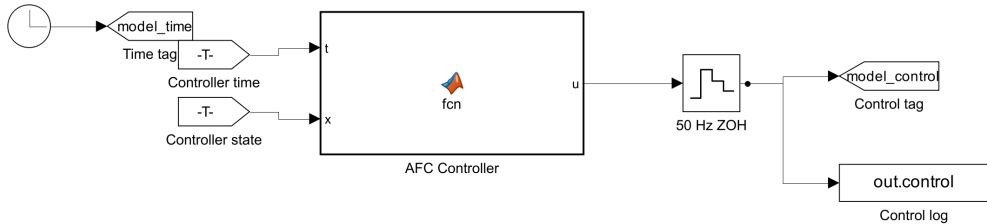
Hardware protocol	50 Hz real-time control, hydraulic valve limited to ± 5 V
Fixed-waveform AFC	body acceleration RMS 1.6651 m/s^2 ; peak 7.21 m/s^2
Mean computation	AFC 4.2 ms; BSC 5.8 ms; PID 2.6 ms
Other road trials	eight sinusoidal conditions, with case-specific gain adjustments for several slower cases

Important: the fixed-waveform hardware acceleration statistics cannot be numerically

compared with our 10-peak sinusoidal simulation.

All values on this slide are paper-reported, not generated by this project.

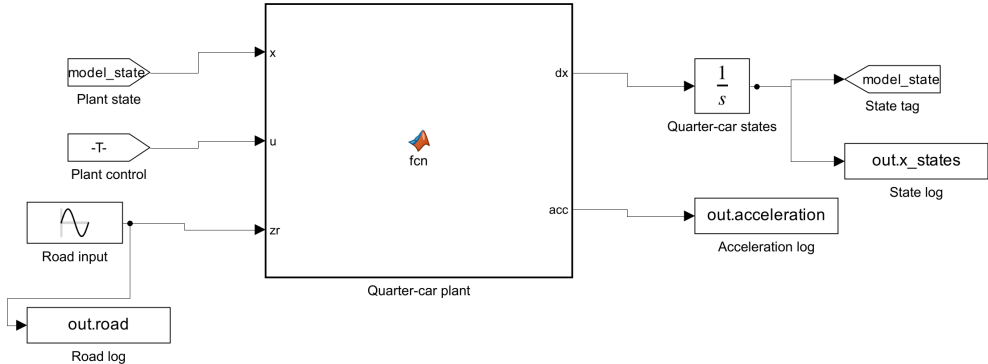
Simulink: sampled AFC and command log



Controller time and state feed the AFC MATLAB Function block; the 50 Hz zero-order hold applies and logs the valve command.

Top crop of the actual executable QuarterCar_AFC_Replication.slx model.

Simulink: quarter-car plant and feedback



The plant computes state derivatives and body acceleration; the integrator closes the feedback loop. Road, states, and acceleration are logged.

Bottom crop of the same Simulink model export.

Parameter ledger: known versus assumed

Quantity	Simulation value	Status
m_s, m_u	320, 40 kg	representative external quarter-car set
k_s, k_t, b_s	18, 200 kN/m; 1 kN s/m	same separate literature set
$A, \beta, P_{\max}$	$3.35 \times 10^{-4} \text{ m}^2$; 1 s^{-1} ; 10.3425 MPa	illustrative actuator values
$\kappa, c_{\text{rel}}, b_u$	10^{-6} , 0.05, 0.75	configurable project assumptions
50 Hz; $\pm 5 \text{ V}$	0.02 s; $\pm 5 \text{ V}$	retained from source paper

The real rig's valve coefficients and raw signals are the main missing inputs for quantitative replication.

Experiment reproduced in MATLAB

Case 10 road

$$z_r(t) = 0.043 \sin(2\pi \cdot 0.505t) \text{ m}$$

20 s horizon, 10 peaks; sample time 0.02 s.

Integration

MATLAB: four RK4 substeps per control interval.

Simulink: fixed-step ode4, 0.001 s.

Comparators

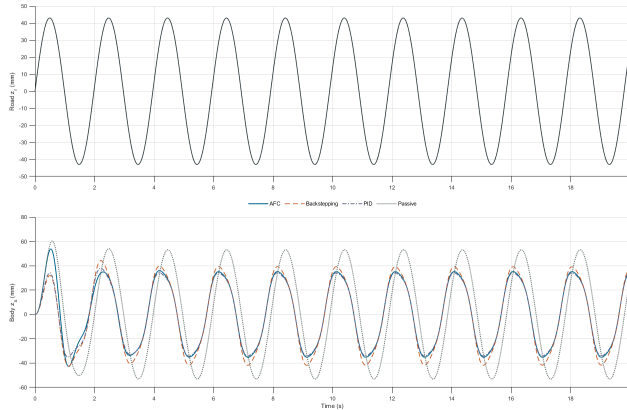
AFC, backstepping, PID, and passive suspension for Case 10.

Diagnostics

IAE, ITAE, ITSE, acceleration RMS, voltage clipping, and PPF violation; AFC sweep across all eight published sinusoidal inputs.

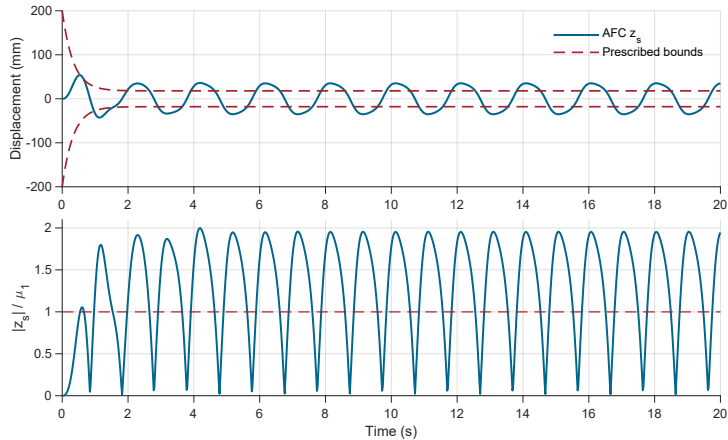
The model uses full simulated state feedback and a surrogate hydraulic pressure-state equation.

MATLAB figure capture: road and body response



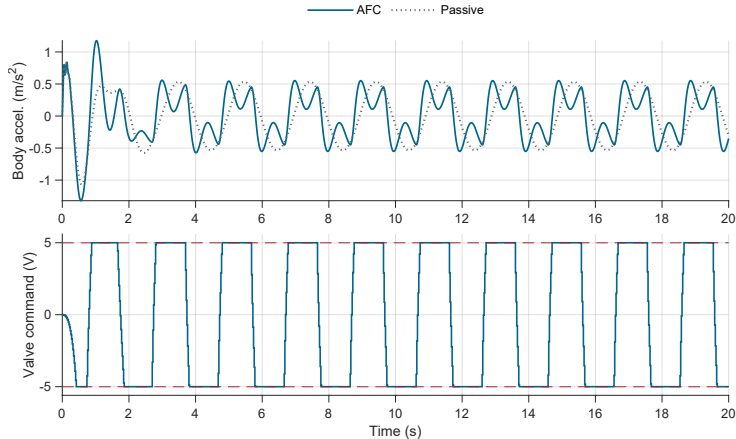
Model output: active control improves displacement relative to passive suspension, but PID narrowly leads AFC under these assumptions.

AFC does not retain the assumed PPF



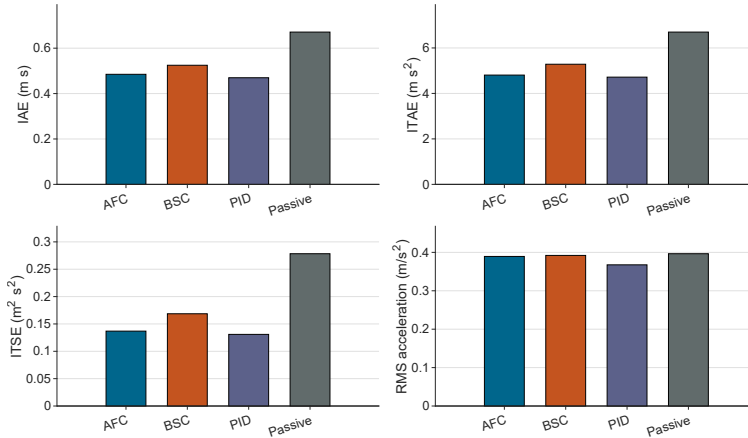
The normalized ratio $|z_s| / \mu_1$ exceeds 1; maximum envelope violation is 0.01885 m.

Actuator authority is the key diagnostic



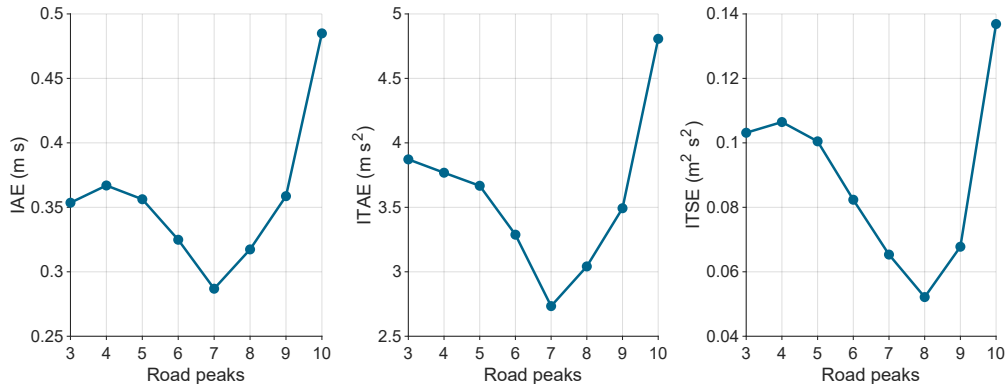
The raw AFC command exceeds the ± 5 V rail in 85.4% of samples; the plant receives the clipped command.

Controller ranking in the assumed model



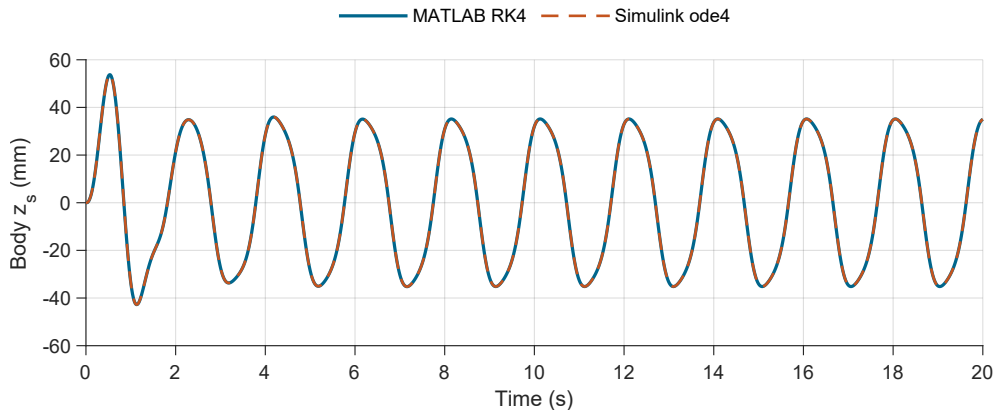
AFC IAE = 0.4849 m s; passive IAE = 0.6709 m s. PID is lower at 0.4698 m s.

Eight published road inputs, fixed AFC tuning



The simulated indices are non-monotonic. The paper retunes several slower cases; this project deliberately holds Case A gains fixed to expose road sensitivity.

Independent numerical implementation check



MATLAB and Simulink agree to a maximum body-displacement difference of 2.10×10^{-8} m for the 10-peak run. This validates code consistency, not physical parameter identification.

Conclusion and next validation step

What is reproduced	AFC recursion, PPF settings, 50 Hz sampled control, sinusoidal road cases, and independent MATLAB/Simulink traces.
What is not reproduced	Original rig/valve identification, fixed-waveform samples, sensor/observer path, and hardware measurements.
What the model reveals	Reduced error versus passive, but severe voltage saturation and PPF violation for Case 10.

Next: identify the hydraulic actuator and obtain the source waveform/time series before claiming quantitative experimental agreement.

Code, model, plots, report, and slides: <https://github.com/Galabavamsi/quarter-car-afc-replication>.

References

-  J. Na et al., “Active Suspension Control of Quarter-Car System With Experimental Validation,” *IEEE TSMC: Systems*, 52(8), 4714–4726, 2022. doi:10.1109/TSMC.2021.3103807.
-  P. Jin, W. Xue, and K. Li, “Actuator Fault Estimation for Vehicle Active Suspensions Based on Adaptive Observer and Genetic Algorithm,” *Shock and Vibration*, 2019. doi:10.1155/2019/1783850.
-  A. K. Abdulzahra and T. Y. Abdalla, “Fuzzy Sliding Mode Control Scheme for Vehicle Active Suspension System Optimized by ABC Algorithm,” *IJISA*, 11(12), 1–10, 2019. doi:10.5815/ijisa.2019.12.01.